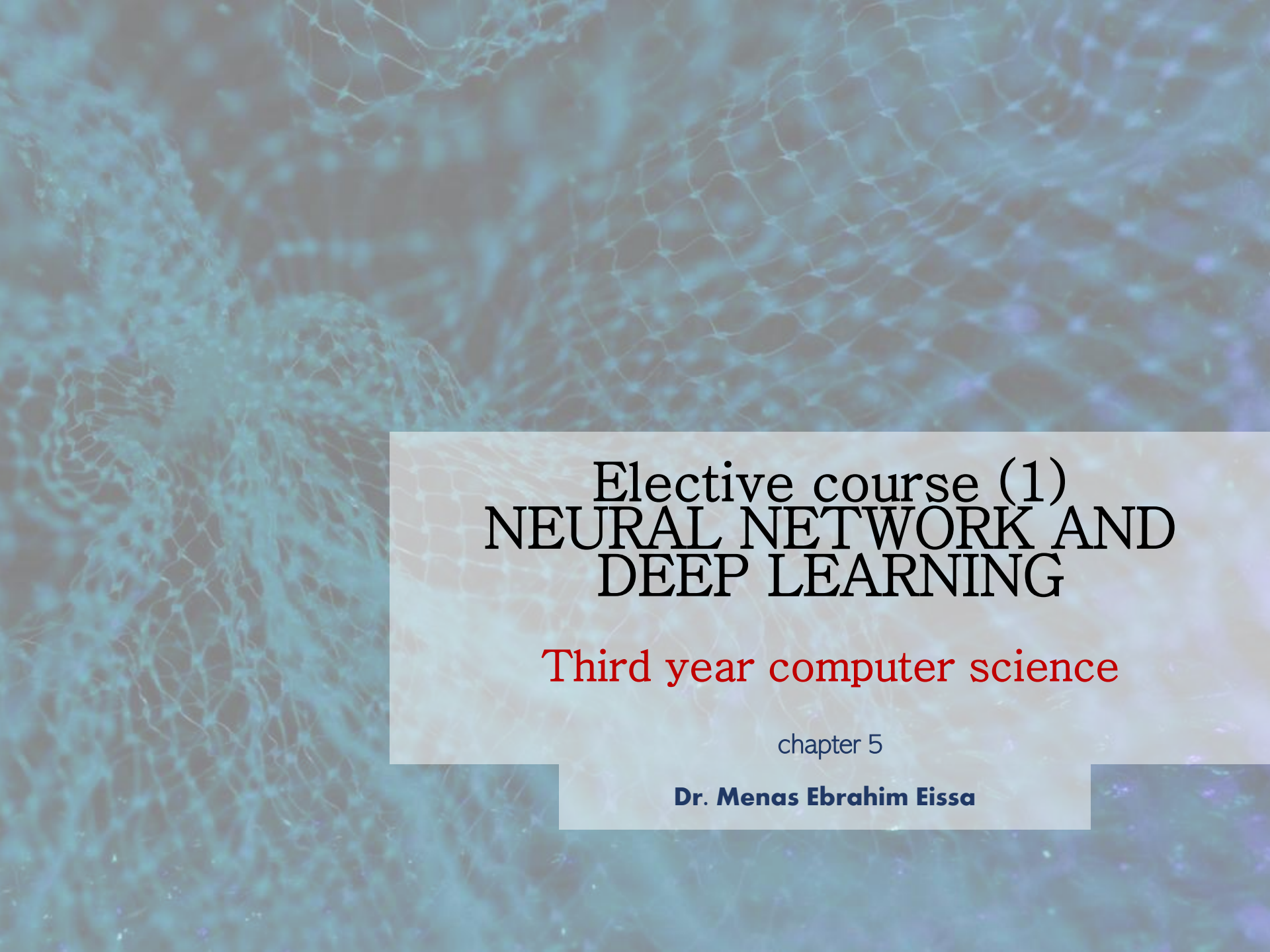




Elective course (1) NEURAL NETWORK AND DEEP LEARNING

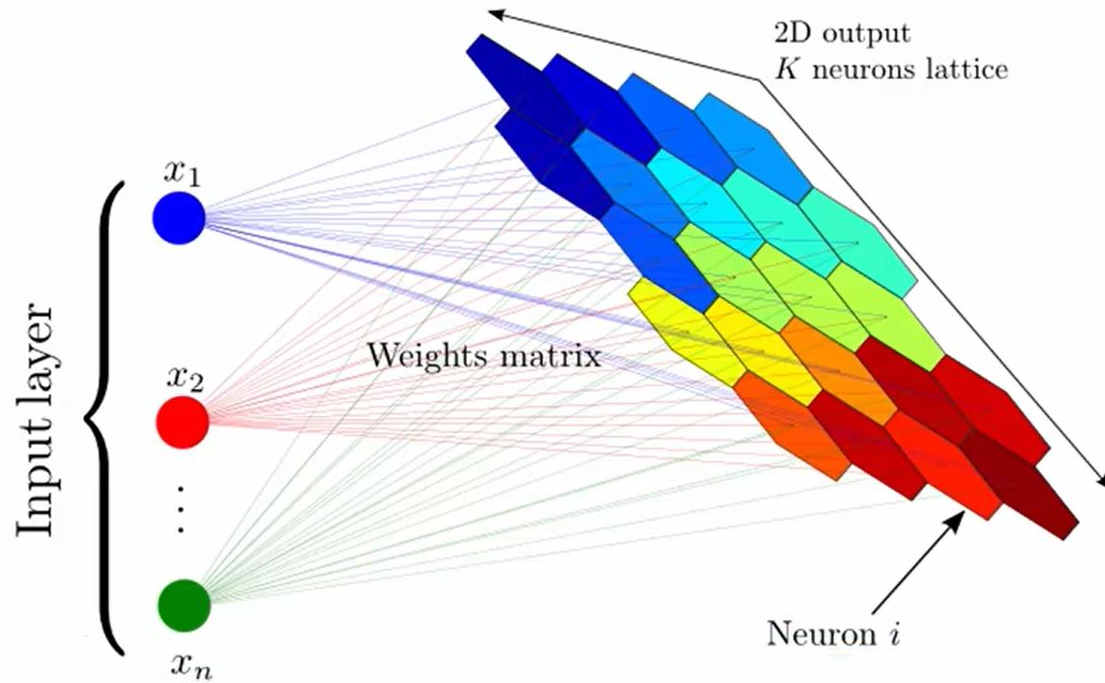
Third year computer science

chapter 5

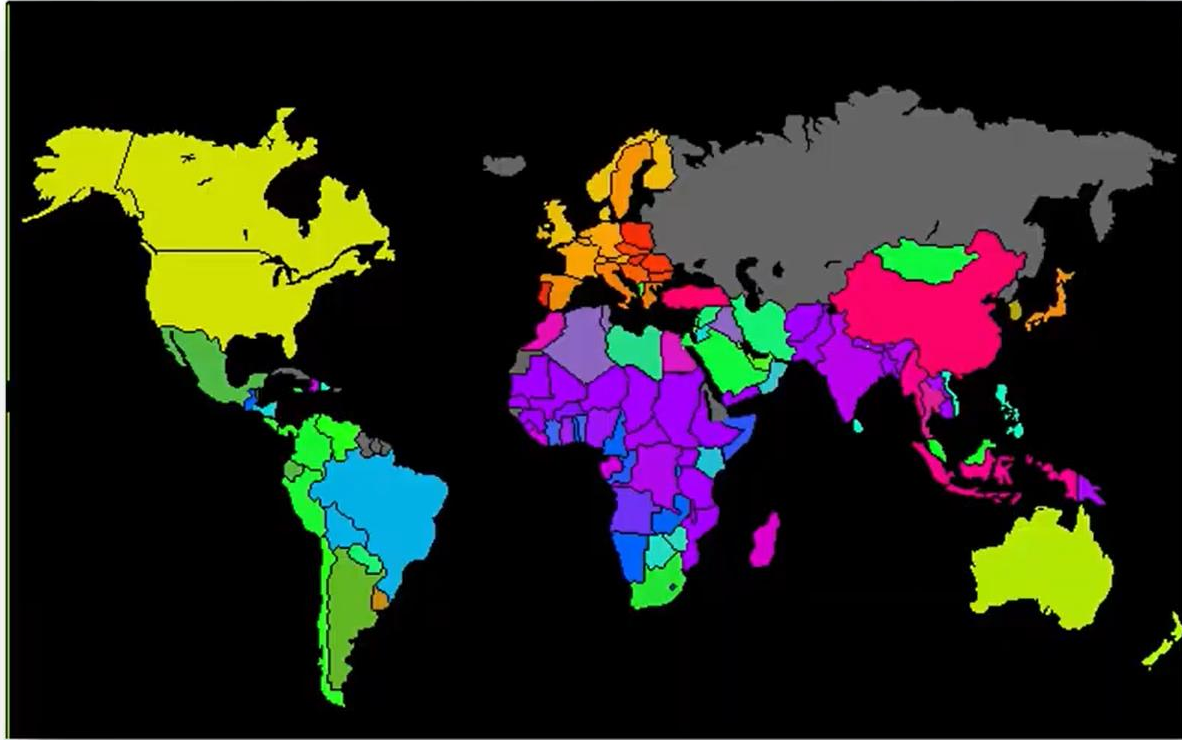
Dr. Menas Ebrahim Eissa

How Do SOMs Work?

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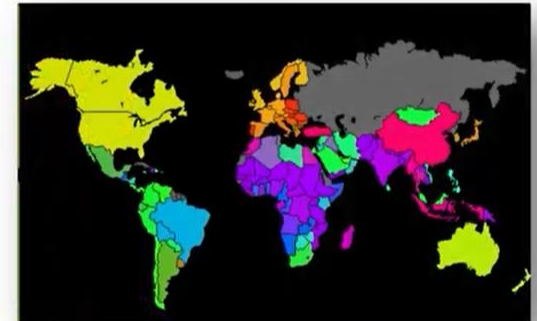
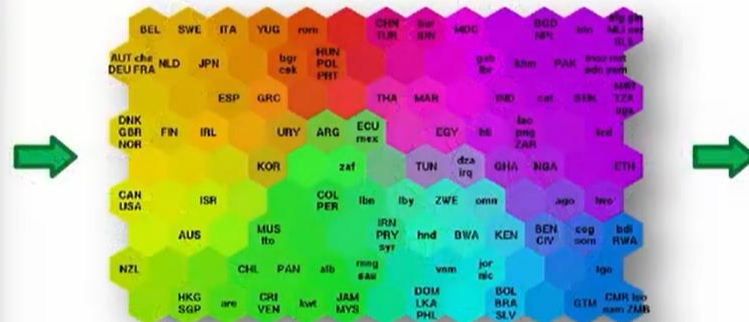


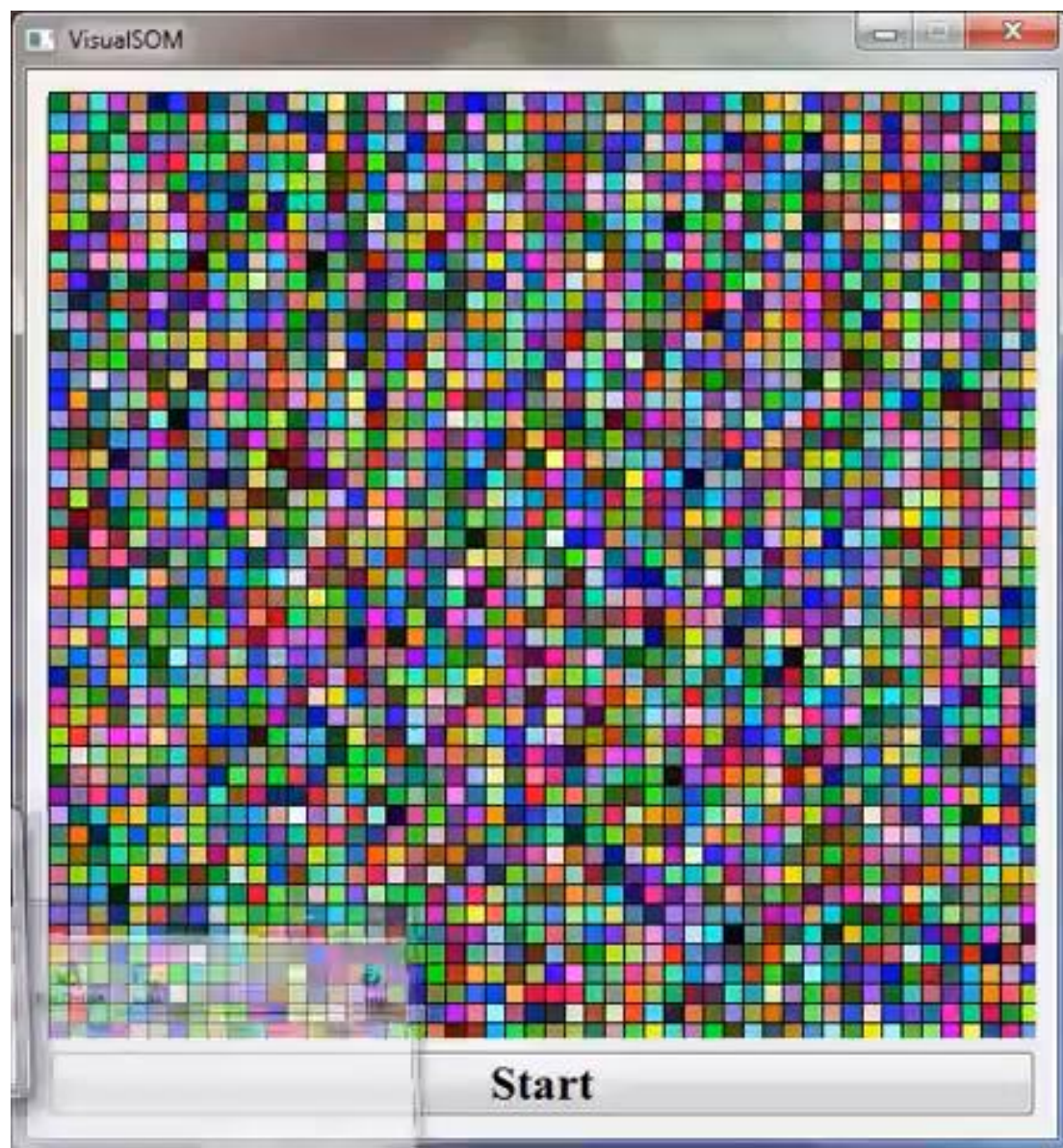
How Do SOMs Work?



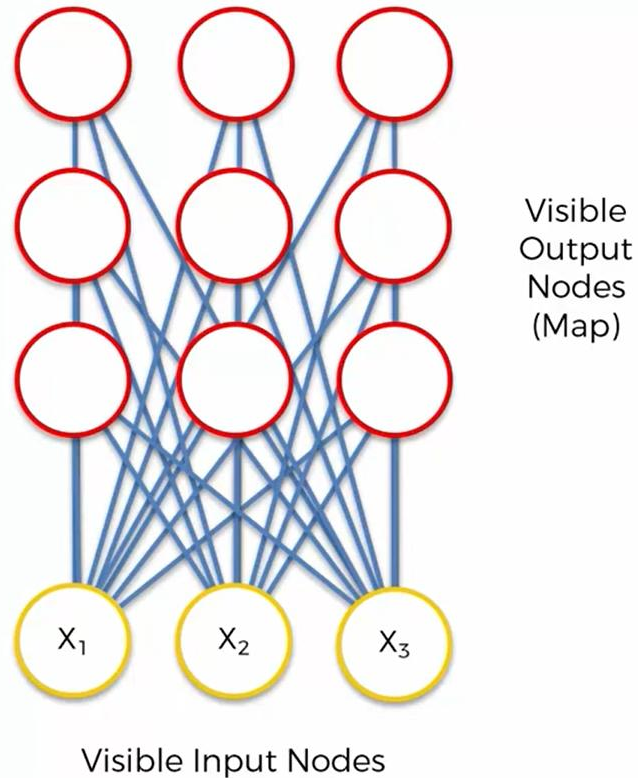
How Do SOMs Work?

	A	B	C	D	E
1	Country	Country C	Health Ex	Education E	Inflation
2	Aruba	ABW	9.418971	5.92467022	-2.13637
3	Afghanistan	AFG	4.371774		-8.28308
4	Angola	AGO	5.791339		13.73145
5	Albania	ALB	6.75969		2.280502
6	Andorra	AND	4.57058	3.1638701	
7	Arab Wor	ARB	4.049924		3.524814
8	United Ar	ARE	7.634758		
9	Argentina	ARG	4.545323	4.88997984	6.282774
10	Armenia	ARM		3.84079003	3.406767
11	American	ASM	4.862062		
12	Antigua a	ATG	9.046056	2.55447006	-0.55016
13	Australia	AUS	11.19444	5.09262991	1.820112
14	Austria	AUT	5.85024	5.7674098	0.506313
15	Azerbaija	AZE	6.964187	3.22430992	1.401056
16	Burundi	BDI	10.39434	6.3197999	10.98147
17	Belgium	BEL	4.46431	6.41535997	-0.05315
18	Benin	BEN	7.405431	4.22204018	2.15683

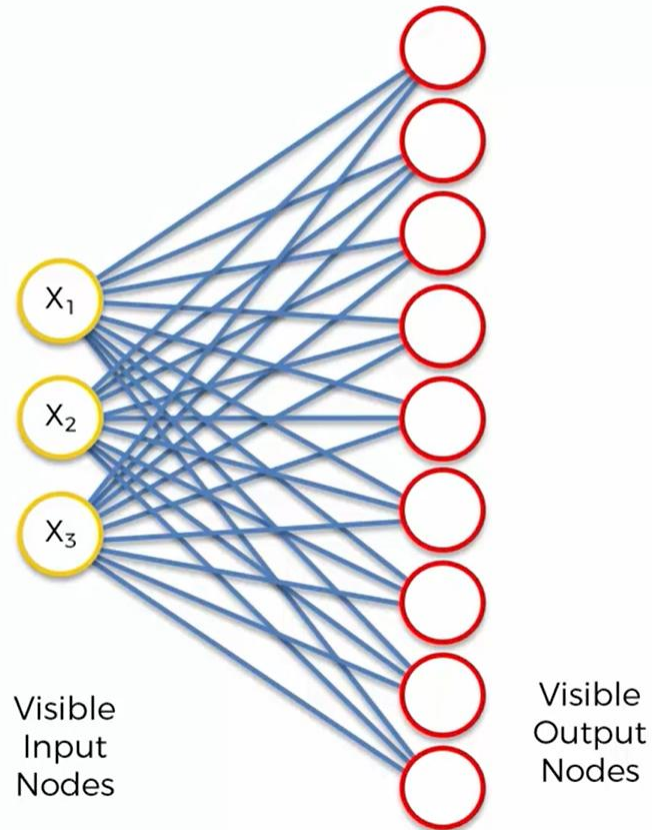




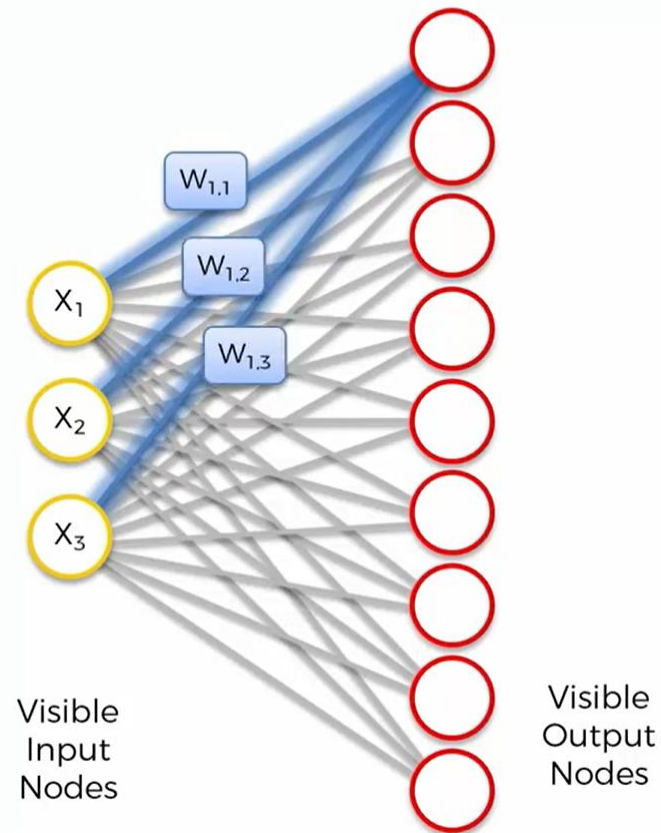
How Do SOMs Learn?



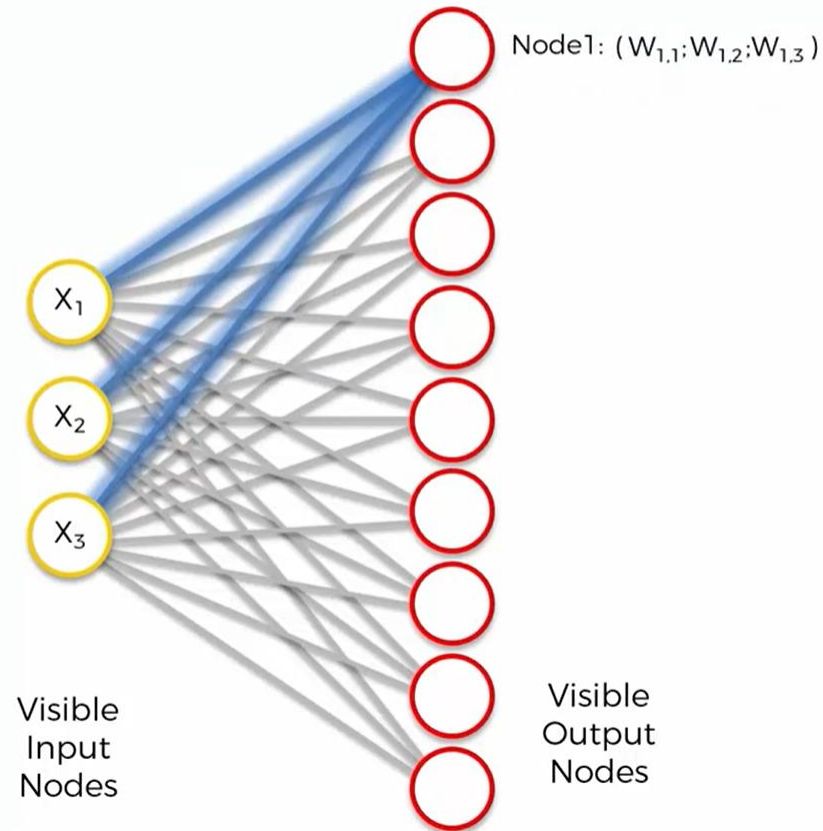
How Do SOMs Learn?



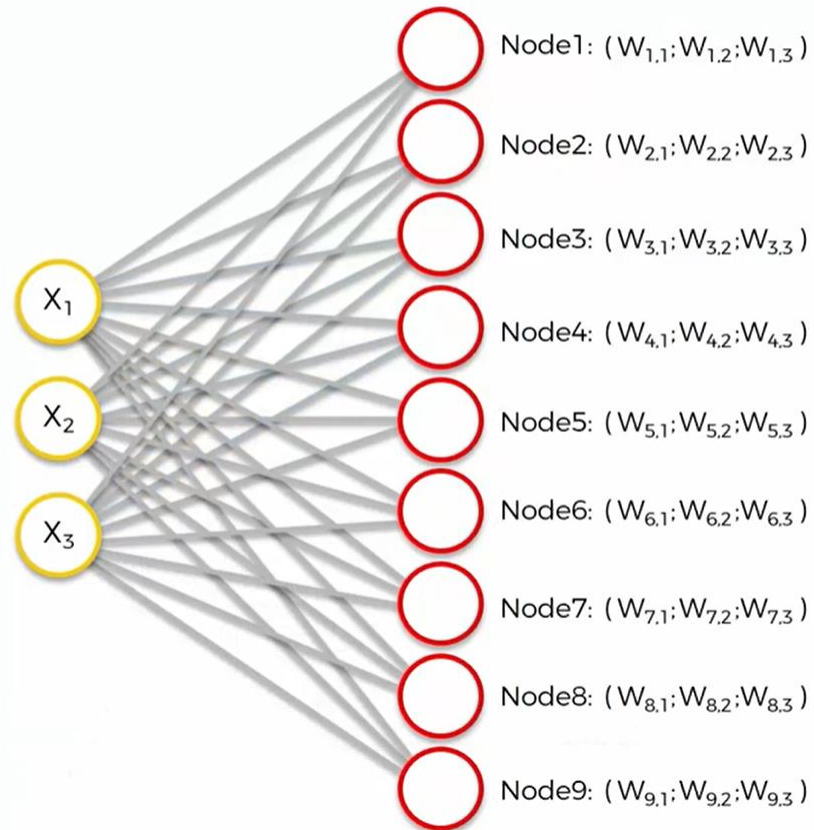
How Do SOMs Learn?



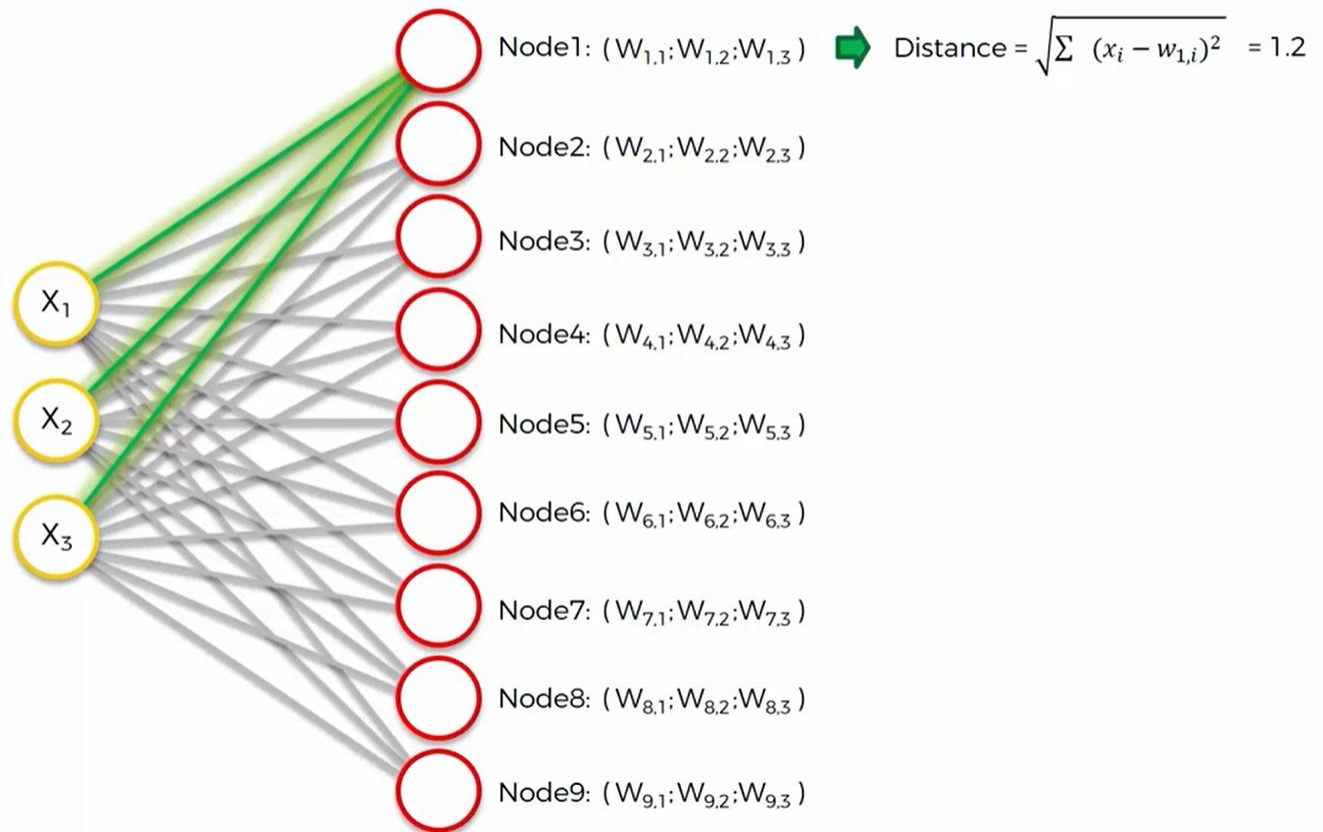
How Do SOMs Learn?



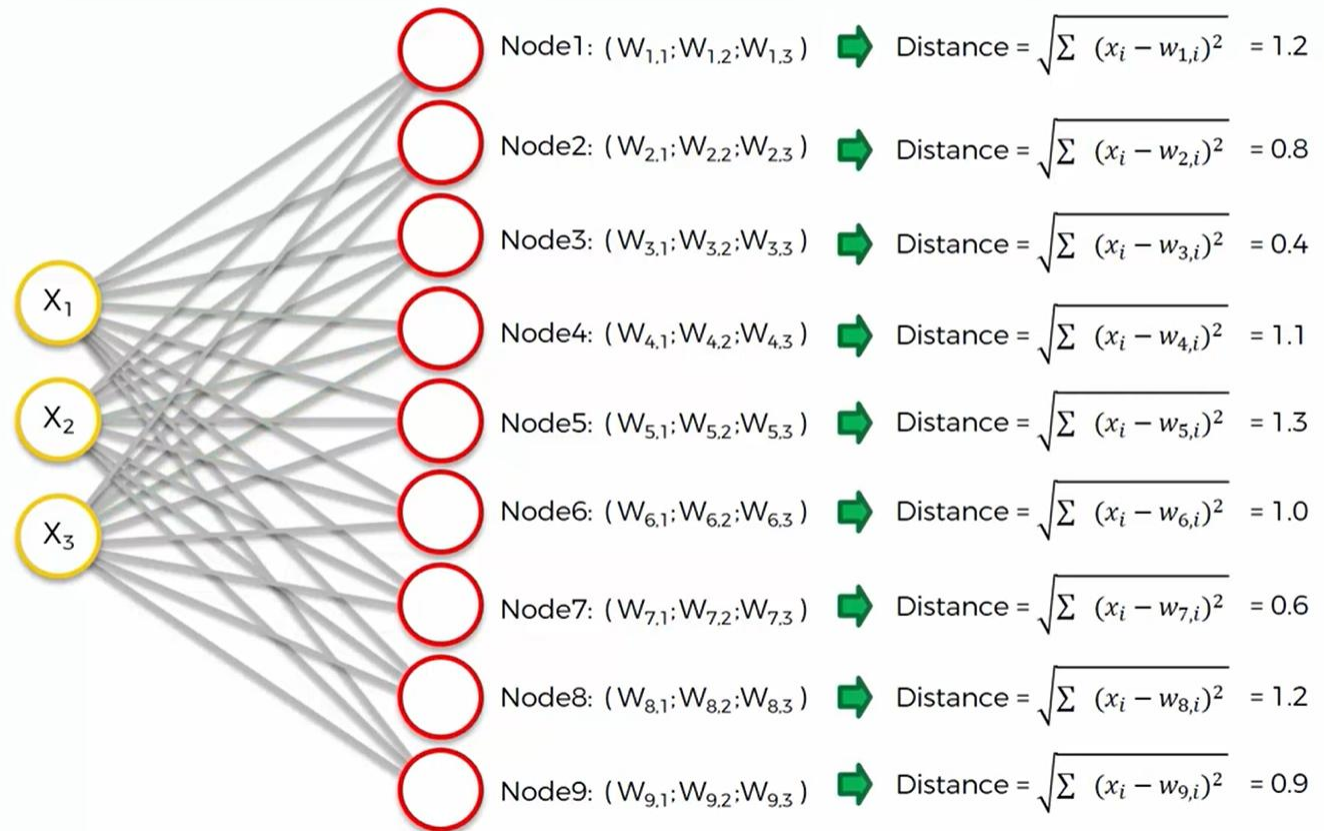
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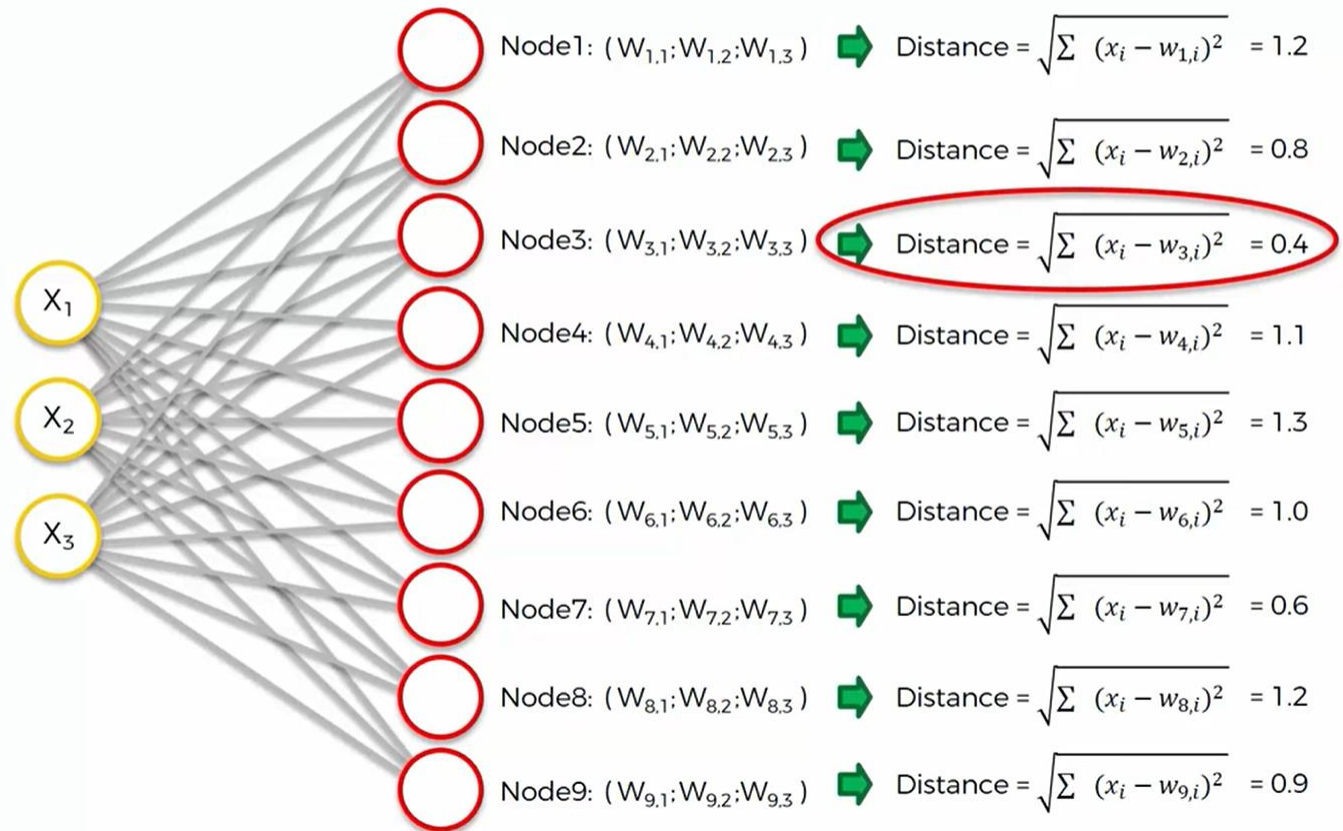
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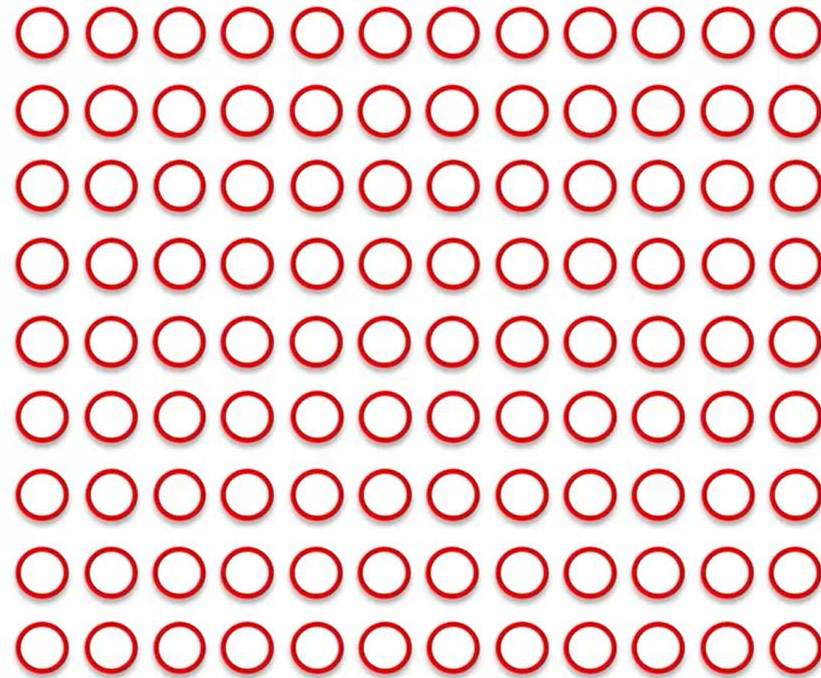
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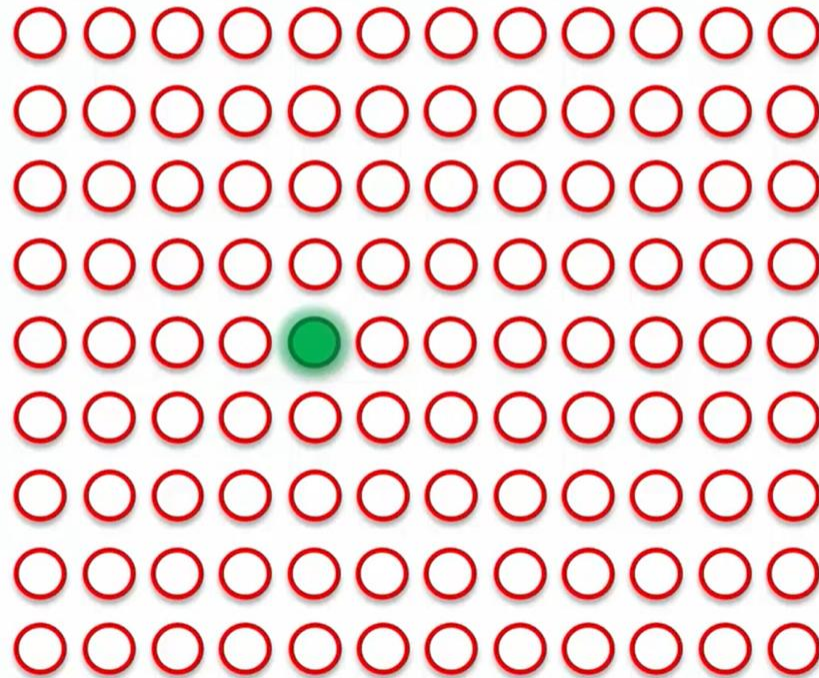
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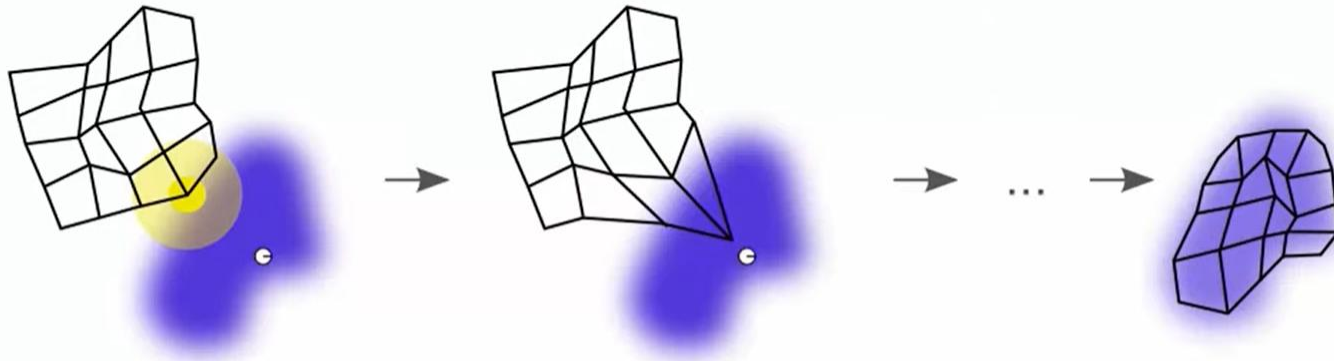
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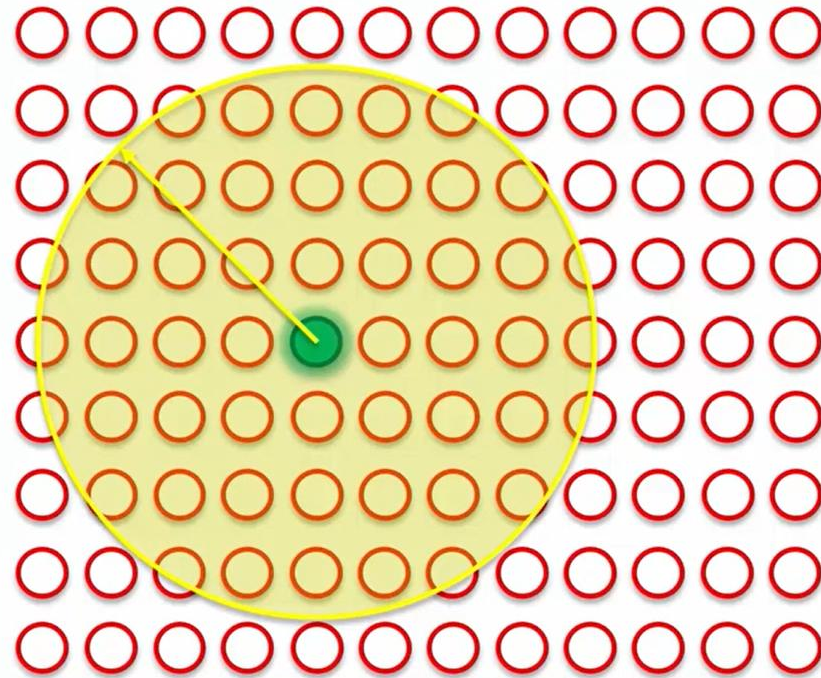
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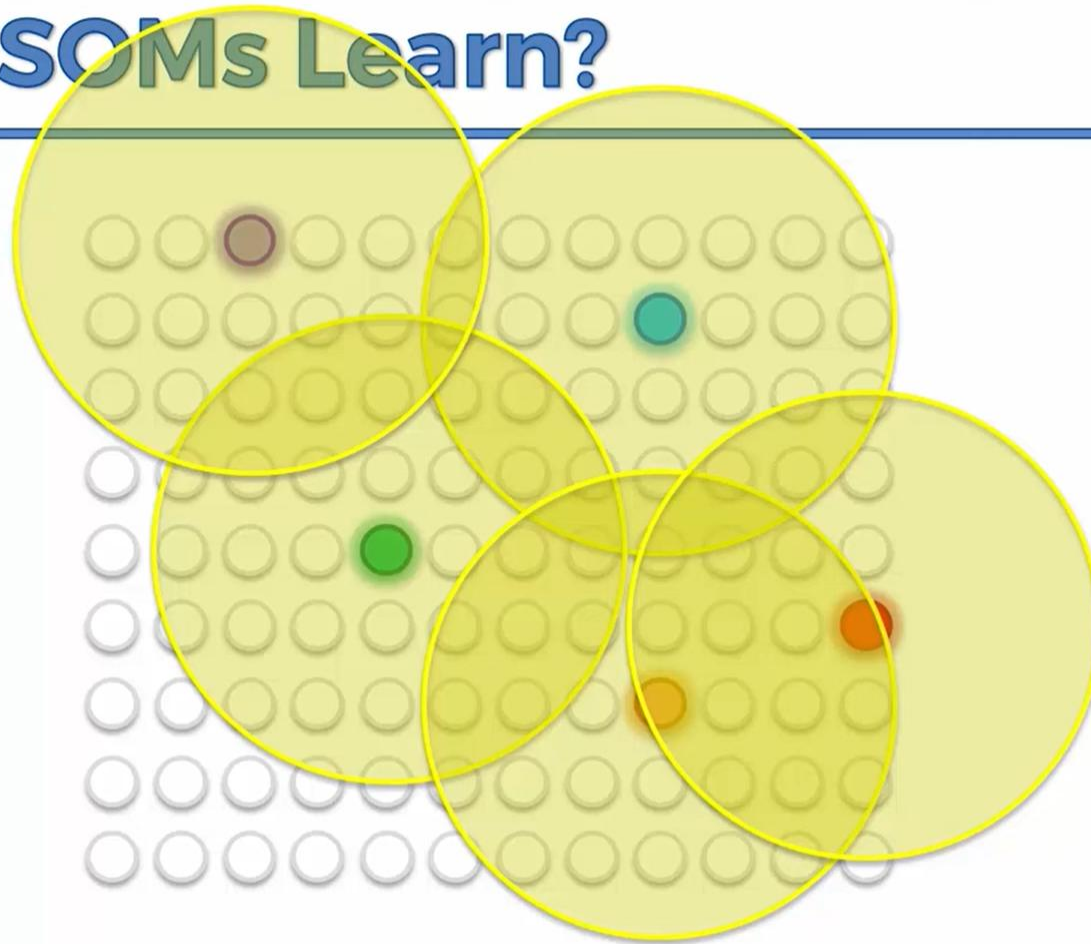
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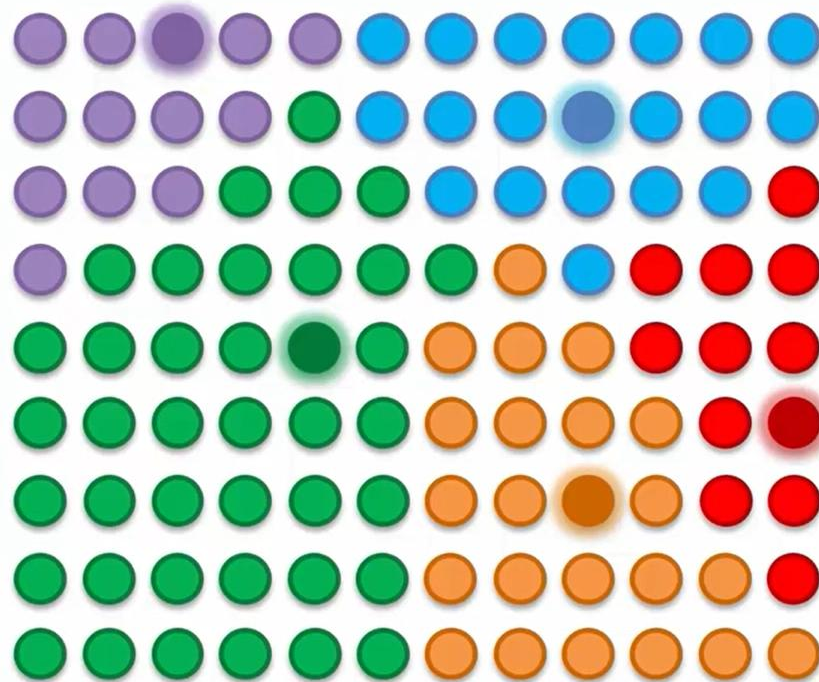
How Do SOMs Learn?



How Do SOMs Learn?



How Do SOMs Learn?



Finding the Best Matching Unit (BMU)

For a given input vector x , the neuron whose weight vector is closest to x is identified as the Best Matching Unit (BMU). This is done by minimizing the Euclidean distance:

$$BMU = \arg \min_i ||x - w_i||$$

Weight Update Rule

The weight vectors are updated using the following rule:

$$w_i(t + 1) = w_i(t) + \eta(t)h_{BMU,i}(t)(x - w_i(t))$$

where:

- $\eta(t)$ is the learning rate, decreasing over time.
- $h_{BMU,i}(t)$ is the neighborhood function, which decreases the effect of updates for neurons farther from the BMU.

Neighborhood Function

A common choice is the Gaussian neighborhood function:

$$h_{BMU,i}(t) = \exp \left(-\frac{\|r_{BMU} - r_i\|^2}{2\sigma^2(t)} \right)$$

where r_{BMU} and r_i are the positions of the BMU and neuron i on the map, and $\sigma(t)$ is the neighborhood radius, which also decreases over time.

Where $\|r_{BMU} - r_i\|^2$ is the Euclidian distance $= \sqrt{\sum (x_i - w_i)^2}$

How Do SOMs Learn?

Important to know:

- SOMs retain topology of the input set
- SOMs reveal correlations that are not easily identified
- SOMs classify data without supervision
- No target vector -> no backpropagation
- No lateral connections between output nodes